

Divyansh Rastogi

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SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning (AI/ML) with hands-on experience in Machine Learning, Deep Learning. Proven ability to build real-world AI systems including transformer-based models (BERT, ELECTRA), Quantum Model(SimCLR) prediction solutions and System Design Project.

EDUCATION

B. Tech in Computer Science and Engineering

Jaypee Institute of Information Technology, Noida

Class XII – CBSE

Brij Bhushan Lal Public School, Bareilly

2023-2027

CGPA: 8.40/10

2021-2022

Percentage: 95.8%

EXPERIENCE

Directorate of Research, Innovation and Development (DRID) | Remote | Project Intern | 2025

- Architected accessible, real-time interactive applications using Python, improving usability for differently-abled users.
- Collaborated in research-driven development, integrating **gamification techniques** to improve stress management engagement.

PROJECTS

Twitter Architecture | System Design

- Engineered a highly scalable Twitter-like microservices architecture handling 10M+ users, comprising 8+ core services (API Gateway, User, Tweet, Timeline, Feed, Search, Media, Notification), achieving 99.9% availability and modular service isolation.
- Incorporated asynchronous processing with Kafka, fan-out workers, and Redis caching to enable low-latency timeline updates and supporting 100K+ writes/sec and 1M+ reads/sec under peak traffic.
- Formulated a scalable data layer using SQL, Cassandra, Elasticsearch, object storage, and graph relationships for users, tweets, search, media, and follower network handling.

Battery Health Prediction using NASA Dataset | ML / DL

- Modeled a time-series battery health prediction system using the NASA dataset, analyzing 500+ charge-discharge cycles to model degradation, real world experience to handle huge datasets(43K Columns) and estimate remaining useful life of EV Battery.
- Performed data preprocessing and feature engineering, including normalization and extraction of key parameters (voltage, current, temperature, State of Charge), improving data consistency by ~25%.
- Configured machine learning and deep learning models (TensorFlow) to predict battery performance and trends with ~90% accuracy.

Emotion Detection using BERT & ELECTRA | DL

- Built an NLP-based emotion classification system using 50K+ Reddit comments, categorizing text into multiple emotional states (7).
- Created an end-to-end pipeline including preprocessing and tokenization, reducing processing time by ~20%.
- Designed a hybrid model integrating a Variational Quantum Circuit (VQC) with transformer embeddings, reducing feature dimensionality and exploring advanced model architecture.

TECHNICAL SKILLS

- **Languages:** C/C++, System Design (Basic)
- **AI/ML:** Machine Learning, Deep Learning, NLP, TensorFlow, Keras, PennyLane, Tokenization.
- **Quantum:** SimCLR, NT-Xent Loss, BERT, ELECTRA

ACHIEVEMENTS & ACTIVITIES

- Solved **400+ Data Structures and Algorithms problems** on coding platforms, strengthening problem-solving
- Volunteered in 20+ technical roles (CICE, AI/ML), supporting events and technical initiatives.
- **Delegate Affairs Team Member, The JOUST(Parola):** Collaborated with a team of 50+ members to drive delegate, contributing to 300+ participant registrations for the annual flagship event.
- **School Exhibition Team Member:** Collaborated with the organizing team to plan and execute the Interschool Science Exhibition, facilitating participation from 30+ teams